Muhammad Ayaan Afzal

Iowa State University Mechanical Engineering (Honors Junior)

mayaan31@iastate.edu (515) 425-2328 3726 Tripp Street mayaanafzal

Profile

Driven mechanical engineering student seeking a summer 2027 co-op/internship in engineering design, manufacturing, or quality assurance.

Work Experience

Manufacturing Engineering Intern 06/2026 – 07/2026 | South Bend, IN
General Stamping & Metalworks
- Designed a CNH Industrial production weld fixture end-to-end - released directly to the line.
- Built a TORO fixture combining poka-yoke error-proofing with an in-process check, and root-caused a part-shifting defect on a separate TORO weld fixture, redesigning clamping and locating to restore weld quality.
- Delivered a weld fixture for customer TMH.
- Drove a Kaizen CIP on an Amada HFA400W bandsaw - a coolant-recapture tray and coolant switch cut weekly loss from 1.5 to 0.5 gal, saving up to $7,600/yr (~$38,000 over 5 years).
- Inspected first-article parts for PPAP using CMM probing software and manual measurements to verify dimensional conformance.
- Contributed to internal Continuous Improvement (CI) projects.

Undergraduate Teacher's Assistant 08/2025 – Present | Ames, IA, US
Mechanics of Materials | Engineering Graphics | Engineering Statics
- Supported instruction for 120 students in stress analysis, buckling, and mechanics of materials.
- Mentored 30 students in CAD, engineering graphics, and GD&T concepts.
- Led recitations for 15 students and graded coursework for a 120-student statics course.

Undergraduate Research Assistant 01/2025 – Present | Ames, IA, US
Mechanical Engineering Department - STAMP Group (Prof. Tuhin Mukherjee)
- Optimize WAAM parameters by simulating droplet transfer and weld pool oscillation.
- Enhance Fortran simulation code to model bulge formation and oscillatory motion.
- Calibrate simulations against thermal and velocity data to reduce defects.
- Conduct parametric studies in TECPLOT 360 on temperature and solidification.

Product Development Engineering Intern 12/2025 – 01/2026 | Karachi, Pakistan
Asia Gas & Electrical Appliances
- Joined the team scaling production of a new electric mini tandoor oven for the German market, working hands-on with die press tooling.
- Redesigned the oven door's bends to standard radii matching existing dies — faster press cycles, no operator misalignment.
- Reverse engineered a thermal appliance, rebuilding its CAD geometry and contributing to the BOM.

Projects

Human-Powered Tiller Design Project (ME 2700) 08/2025 – 12/2025
- Directed SolidWorks design and material selection for a human-powered tiller.
- Ran DFM/DFA and tracked the BOM to hit a $150 budget.
- Fabricated a functional prototype on the lathe, mill, bandsaw, and drill press.

Makala Soprano Ukulele (Model MK-S) 07/2025 – 08/2025
- Recreated a Makala soprano ukulele from direct measurement.
- Modeled a high-fidelity CAD assembly in SolidWorks with complex surfaces.
- Printed a 3D scale replica matching the original geometry.

Tire Rim Design for Solar Car — PRISUM Team 06/2025 – 07/2025
- Produced a lightweight rim via 3D printing, engineered for strength and cost.
- Tuned geometry for real-world driving loads and vehicle dynamics.
- Balanced aesthetic and functional requirements for the PRISM team.

Piston Engine – 4-Cylinder Assembly 06/2025 – 06/2025
- Assembled a 4-cylinder piston engine from scratch in SolidWorks.
- Simulated crankshaft and piston motion using mechanical mates.
- Strengthened assembly design and constraint-management skills.

Education

Bachelor of Science in Mechanical Engineering (Honors) 05/2028 | Ames, IA, U.S
GPA: 3.86/4.0

Skills

SOLIDWORKS | Tecplot 360 | Computer-Aided Design (CAD) | Python | Fortran | MATLAB | MS Excel

Honors and Awards

Editor's Choice Article 02/2026
Machines (MDPI)
"A Review on In-Situ Monitoring in Wire Arc Additive Manufacturing: Technologies, Applications, Challenges, and Needs"

First Place – American Welding Society (AWS) Poster Competition (Undergraduate, B.S. Engineering)
Achieved 1st place out of 25 student entries and secured a $750 monetary award.